Siltwatch

Siltwatch: Executive Summary

Problem

Street flooding in small towns is usually caused by something mundane: a catch basin packed with sediment, leaves, and trash that can no longer take in water. Towns with fewer than 25,000 residents typically have between 300 and 1,500 catch basins and one or two public works employees available to maintain them. Because nobody can see inside a basin without lifting the grate, crews work from a fixed calendar, usually one pass per year, or respond after a resident calls to report standing water. Both approaches fail in predictable ways. Calendar cleaning spends labor on basins that are mostly clear, while the few that fill quickly, often those downhill from construction or unpaved roads, clog between visits. Reactive cleaning means the flood has already happened, along with the property damage, the emergency overtime, and the complaints to the town council.

In interviews with 14 stormwater and public works managers in three states, 12 told us that an inability to know which drains need attention is their primary maintenance problem. Eleven said they had no budget for a full asset-management system, but all 14 said a low-cost tool that pointed crews to the right drains would change how they schedule work.

Solution

Siltwatch is a solar-powered sensor and phone app. The sensor mounts under the grate inside a catch basin. An ultrasonic ranging module measures the distance to the sediment surface and to the water surface every 15 minutes. A small solar panel on the grate frame charges a battery sized to run through two weeks of overcast weather, and the unit reports over a low-power cellular connection that we manage, so towns do not need their own network or IT staff.

The app turns raw readings into a ranked list. Each instrumented basin receives a clog-risk score based on sediment depth, the rate at which sediment is accumulating, and how slowly water drains after rain. Crews open the app each morning and see the ten basins most in need of cleaning. The app sends an alert when a basin crosses a threshold or when heavy rain is forecast and a high-risk basin is still uncleaned. Each cleaning is logged with a timestamp and a photo, which produces the maintenance records that towns must keep under their stormwater permits.

Our design priorities are low cost, no installation tools beyond a wrench, and a crew member learning the app in under ten minutes. Our current prototype unit costs $62 to build in batches of 100, and we project $41 at a volume of 1,000.

Market

There are roughly 19,000 incorporated places in the United States with populations under 25,000, and most are responsible for their own stormwater systems under federal permit requirements. We estimate an average of 600 basins per town. Instrumenting even the highest-risk 15 percent, about 90 basins per town, represents a serviceable market of roughly 1.7 million sensors. At our planned pricing, that is a long-term opportunity of about $400 million in annual software revenue. Our initial focus is narrower: towns in the Midwest and Northeast with populations between 3,000 and 20,000, a group of approximately 4,200 places where flooding from clogged drains is a recurring council issue.

Business Model

We sell each sensor for $149 and charge a software subscription of $60 per sensor per year, which covers cellular data, cloud hosting, alerts, and compliance reports. A typical first purchase is 40 sensors, or $5,960 in hardware plus $2,400 annually. Many towns can fund this from a discretionary maintenance line without a bid process, which shortens the sales cycle. At that pricing, hardware gross margin is 58 percent at current build cost and rises to about 72

percent at volume, while software margin is above 80 percent. We expect towns to expand coverage after the first season, so subscription revenue per customer should grow without additional selling cost.

Competition

Large water-technology vendors sell level sensors and monitoring platforms aimed at city utilities, at prices from $800 to $2,000 per node plus integration fees, which puts them out of reach for our target towns. Some cities rely on smart-lid or camera-inspection services, which are priced per visit and do not give continuous readings. Our real competitor is the status quo: the annual calendar and the resident phone call. Our advantage is price, simplicity, and focus on one decision, which basin to clean next. Our weakness is that we are new, our sensor has been tested for months rather than years, and an established vendor could lower prices if we prove the market. We believe our small-town focus and low support burden are hard for large vendors to match.

Team

Maya Okonkwo-Reyes, CEO, is a senior in civil and environmental engineering. She spent a summer interning with a county stormwater division and conducted most of our customer interviews. Daniel Trujillo, CTO, is a senior in electrical engineering who designed the sensor board and power system. Priya Venkataraghavan, head of software, is a junior in computer science who built the app and the risk-scoring model. All three work on Siltwatch part time alongside coursework. We lack a founder with municipal sales experience, and recruiting an advisor with that background is our top priority.

Milestones and Financials

Siltwatch LLC was formed this year. Our core concept was defined after the interview phase, and we have since built three hardware revisions. We completed a pilot in two municipalities with 38 sensors installed for a full storm season. The sensors flagged 9 basins for early cleaning before the heaviest rainfall; crews cleaned them, and neither town reported street flooding at those sites. Sediment readings matched manual measurements within 1.5 centimeters in 31 spot checks. One sensor failed due to a loose connector, which we fixed in the next revision. Both pilot towns have told us they intend to buy, though neither has signed a purchase order, and this is our largest gap.

To date we have spent $11,400, funded by $6,000 in founder contributions and a $5,400 university innovation grant. Over the next 12 months we project 6 paying towns and 240 sensors, generating about $50,000 in revenue against $118,000 in costs, with break-even expected in the third year as customers expand coverage.

What We Are Asking For

We are seeking $40,000 in prize funding. About $22,000 would pay for a first production run of 200 sensors and enclosure tooling, $10,000 for field certification and outdoor durability testing through a full winter, and $8,000 for sales travel and conference attendance to reach municipal buyers. Beyond funding, we are asking for introductions to experienced municipal technology advisors and mentors who can help us convert our two pilot towns into signed contracts and design a repeatable sales process.